

2. The current, I , in a wire of cross-sectional area, A , is given by the equation:

$$I = nAve$$

- (a) (i) State what the letter n represents in this equation. [1]

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- (ii) Using a labelled diagram, derive the equation above. [4]

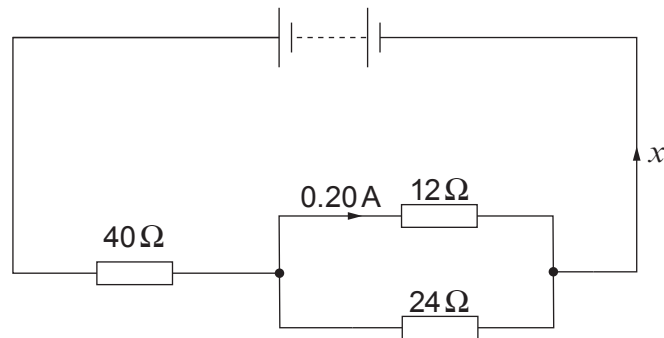
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- (iii) In a physics experiment there is a current of 1.8 A in a nichrome wire of diameter 0.19 mm. For nichrome, $n = 9.0 \times 10^{28} \text{ m}^{-3}$. Calculate the drift velocity. [2]

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- (b) The nichrome wire is cut into different lengths to make three resistors. The resistors are connected in the circuit as shown. The battery has negligible internal resistance.



- (i) Explain, in clear steps, why the current, x , is $0.30\ \text{A}$. [3]

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- (ii) Calculate the pd across the supply. [3]

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- (iii) Show that the power dissipated by the $12\ \Omega$ and $24\ \Omega$ parallel arrangement is **one fifth** that dissipated by the $40\ \Omega$ resistor. [3]

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